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Los de volgende complexe vierkantsvergelijking op in $\mathbb{C}$:

$$z^2 = (1 - i)(z - 2)$$

$$\Leftrightarrow z^2 = z - 2 - zi + 2i$$

$$\Leftrightarrow z^2 - (1 - i)z + (2 - 2i) = 0$$

$$\Delta = (1 - i)^2 - 4(2 - 2i) = (1 - i - i + i^2) - 8 + 8i = -8 + 6i$$

Wortel van discriminant berekenen:

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -8 \\ 2xy = 6 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -8 \\ xy = 3 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = 8 \\ x^2(-y)^2 = 9 \end{cases}$$

x^2 en y^2 zijn oplossingen van:

$$\lambda^2 + 8\lambda - 9 = 0$$

$$\lambda_{1,2} = \frac{-8 \pm \sqrt{100}}{2}$$

$$\lambda_1 = 1 \text{ en } \lambda_2 = -9$$

$$x = 1 \text{ en } y = 3$$

Oplossing wortel discriminant: $1 + 3i$ of $-1 - 3i$

$$z_1 = \frac{1 - i + 1 + 3i}{2} = 1 + i$$

$$z_2 = \frac{1 - i - 1 - 3i}{2} = -2i$$

References